

1. Which of the following is an SI base unit?
 - (A) Ampere
 - (B) Volt
 - (C) Ohm
 - (D) Coulomb

2. 3.1415926 expressed to TWO significant figures is
 - (A) 3.1
 - (B) 3.14
 - (C) 3.2
 - (D) 31

3. When used in front of a unit the prefix 'mega' means
 - (A) 10^{-6}
 - (B) 10^{-3}
 - (C) 10^3
 - (D) 10^6

4. The time period of a simple pendulum vibrating with a small amplitude depends on the
 - (A) mass of the pendulum bob
 - (B) amplitude of the oscillation
 - (C) length of the pendulum
 - (D) force with which the pendulum is set into motion

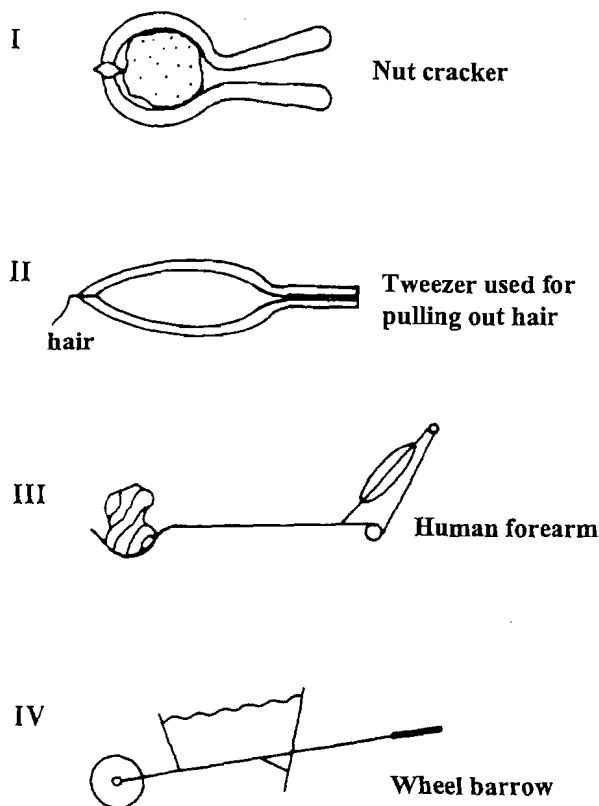
5. A slice of bread is squeezed into a little ball. Which quantity does NOT change?
 - (A) Mass
 - (B) Volume
 - (C) Density
 - (D) Width

6. The relative density of brass is 8.4. Which of the following is the BEST statement that can be made?
 - (A) Brass is heavier than water.
 - (B) Water is heavier than brass.
 - (C) 100 cm^3 of brass is heavier than 100 cm^3 of water.
 - (D) 100 cm^3 of water is heavier than 100 cm^3 of brass.

7. The force which keeps the moon in orbit is
 - (A) magnetic
 - (B) centrifugal
 - (C) reactive
 - (D) gravitational

8. Which of the following, when displaced, is seen to be in neutral equilibrium?
 - (A) A heavy weight suspended on a string
 - (B) A pencil balanced on its point
 - (C) A beam of a balance in use
 - (D) A cone resting on a slanted edge

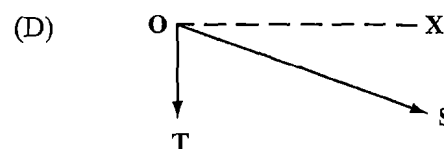
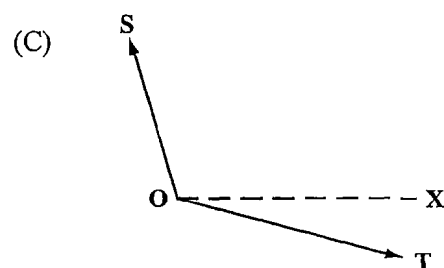
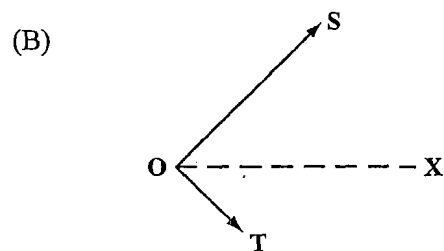
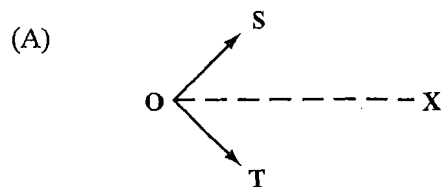
Item 9 refers to the following diagrams which depict some simple machines.



9. In which of these machines would the effort applied be LESS than the load?

- (A) I and II only
- (B) I and IV only
- (C) II and III only
- (D) II and IV only

10. The diagrams below, drawn to scale, represent two forces S and T acting at O. In which of the following is the resultant in the direction, OX?

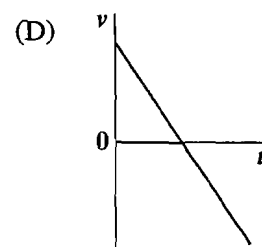
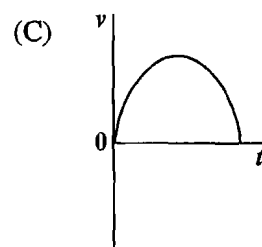
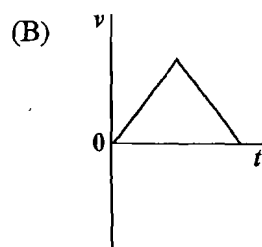
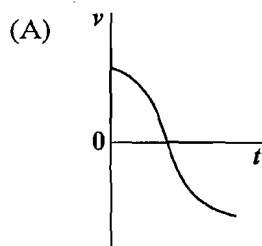


11. On which of the following does the pressure at a point in a liquid depend?

- I. Density of the liquid
- II. Depth from the surface
- III. Area of the cross-section of the container

- (A) I only
- (B) III only
- (C) I and II only
- (D) I, II and III

12. Which of the following speed-time graphs could represent the motion of a cricket ball which is thrown vertically upwards and caught by the thrower?



13. A glass marble, X, moving with a speed of 6 ms^{-1} , collides 'head on' with an identical stationary glass marble, Y. What is the velocity of Y after collision assuming that X is brought to rest?

- (A) 0 ms^{-1}
 (B) 3 ms^{-1}
 (C) 6 ms^{-1}
 (D) 12 ms^{-1}

14. When an object is floating on a liquid, which of the following would be true?

- I. The object weighs the same as the volume of water displaced.
 II. The upthrust on the object is slightly greater than the weight of the object.
 III. The average density of the object is just equal to that of the liquid.

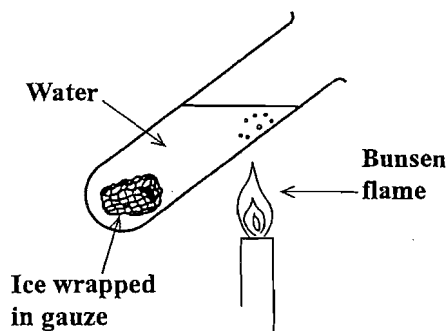
- (A) I only
 (B) III only
 (C) I and II only
 (D) I, II and III

15. A book of weight 12 N is resting on a table. The force that the table exerts on the book is

- (A) 0 N
 (B) 12 N
 (C) greater than 12 N
 (D) greater than 0 N but less than 12 N

16. A particle moving with constant speed in a uniform circular path, has a
- (A) constant velocity
 - (B) constant acceleration along the radius of its path
 - (C) variable acceleration
 - (D) constant acceleration tangential to its path
17. Which of the following is NOT true about the moment of a force?
- (A) It is measured in units called newton metres.
 - (B) It is the amount of force needed to keep a body turning.
 - (C) It is the turning effect that a force has when it acts on a body.
 - (D) It is the product of the force and the perpendicular distance of its line of action from a point.
18. Which of the following is/are renewable source(s) of energy?
- I. Solar energy
 - II. Wind energy
 - III. Geothermal energy
- (A) I only
 - (B) I and III only
 - (C) II and III only
 - (D) I, II and III
19. Which of the following implements is/are designed to take advantage of a large moment provided by a relatively small force?
- I. Claw hammer
 - II. Crowbar
 - III. Pair of tweezers
 - IV. Pair of wire cutters
- (A) III only
 - (B) I and IV only
 - (C) I, II and III only
 - (D) I, II and IV only
20. Which of the following would be characteristic features of a clinical thermometer?
- I. Narrow constriction in the tube just above the bulb
 - II. Limited range of temperatures
 - III. Large bore to make the mercury more visible
- (A) I only
 - (B) I and II only
 - (C) I and III only
 - (D) I, II and III

Item 21 refers to the diagram below which shows water boiling at the top of a glass test tube while a piece of ice remains unmelted at the bottom.



21. Which of the following is a reason for this occurrence?

(A) Water is a poor conductor of heat.
 (B) Gauze is a poor conductor of heat.
 (C) Water is a good conductor of heat.
 (D) Glass is a good conductor of heat.

22. Which of the following types of radiation is responsible for an atom of $^{14}_6\text{C}$ changing into an atom of $^{14}_7\text{N}$?

(A) α
 (B) β
 (C) γ
 (D) n

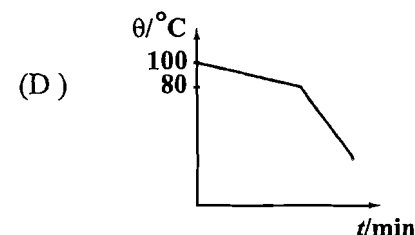
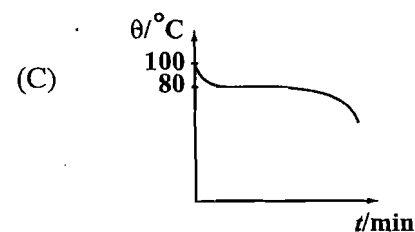
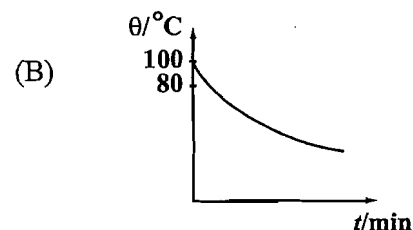
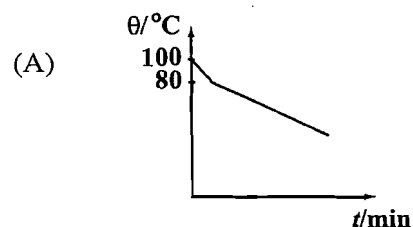
23. A flask contains air under pressure. Some of the air is let out slowly over a period of 10 seconds. When the flask is closed the

(A) pressure of the air in the flask will have increased
 (B) volume of air in the flask will have decreased
 (C) temperature in the flask will have increased
 (D) number of molecules striking the wall per second will have decreased

24. The heat capacity of a body depends on

(A) its mass and the specific heat capacity of the material
 (B) its mass and the temperature change the body undergoes
 (C) the density and temperature of the material
 (D) the density and expansivity of the material

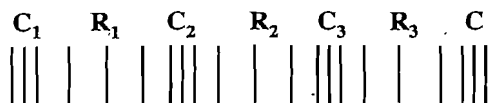
25. Some molten naphthalene at 100°C is allowed to cool down at room temperature. If naphthalene has a melting point of 80°C , which of the following BEST represents the cooling curve?



26. An electric kettle full of water is plugged into the mains. The process by which heat travels through the water is

(A) electrification
(B) evaporation
(C) convection
(D) radiation

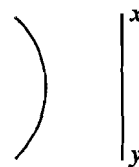
Item 27 refers to the following diagram which shows a longitudinal wave, where, C and R represent compressions and rarefactions respectively.



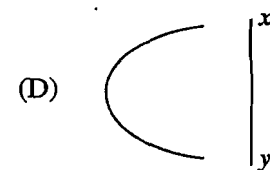
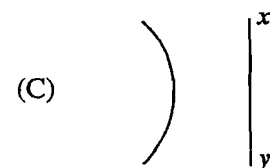
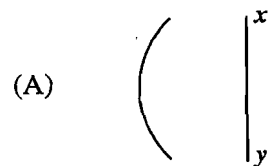
27. If λ = wavelength, the distance between C_1 and R_2 is

(A) $\frac{1}{2} \lambda$
(B) 1λ
(C) $1\frac{1}{2} \lambda$
(D) 2λ

Item 28 refers to the following diagram.

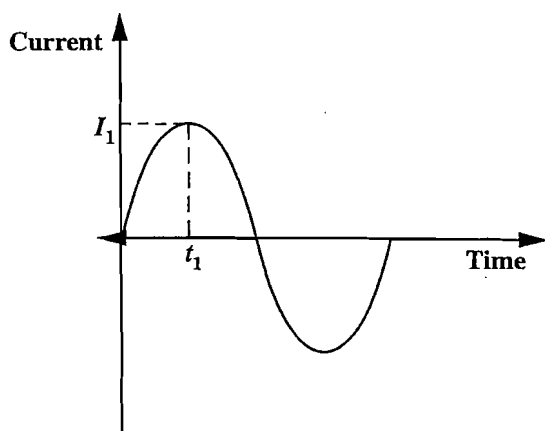


28. The circular wavefront shown above is progressing towards the reflecting surface xy . Which of the diagrams below correctly shows the shape of the wavefront after reflection?



29. Which of the following statements about sound waves, water waves and electromagnetic waves is NOT correct?
- (A) They all transmit energy from the source to the surroundings.
 - (B) They are all examples of transverse waves.
 - (C) They all undergo reflection, refraction and diffraction.
 - (D) They all have different speeds of propagation.
30. In which of the following are the electromagnetic waves written in order of INCREASING frequency, starting with the lowest?
- (A) X-rays, ultraviolet, infrared, radio
 - (B) X-rays, infrared, ultraviolet, radio
 - (C) Radio, x-rays, infrared, ultraviolet
 - (D) Radio, infrared, ultraviolet, x-rays
31. Which of the following demonstrates the straight line propagation of light?
- (A) The spreading of light when it passes through a narrow opening
 - (B) The dispersion of light by a prism
 - (C) A parallel beam of light coming to a point focus after reflection from a spherical mirror
 - (D) The operation of the pinhole camera
32. The image formed in a plane mirror is
- (A) real and upright
 - (B) real and inverted (upside down)
 - (C) virtual and upright
 - (D) virtual and inverted (upside down)
33. A ray of light in air strikes a glass block at an angle of incidence of 0° . The light will be
- (A) undeviated
 - (B) totally reflected
 - (C) refracted at 90° to normal
 - (D) refracted at an unknown angle
34. Which of the following statements about waves is true?
- (A) Only transverse waves undergo reflection.
 - (B) All waves undergo reflection, refraction and diffraction.
 - (C) Diffraction can only take place with light waves.
 - (D) Longitudinal waves do not undergo reflection, but may be reflected.

Item 38 refers to the graph below which shows the variation of alternating current with time.



38. The value of the current I_1 at time t_1 is called the

(A) sinusoidal value
(B) peak value
(C) root mean square value
(D) average value

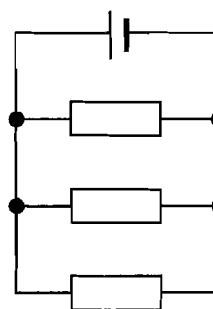
39. Which of the following equations CANNOT be used to determine the power dissipated in a resistor?

(A) $P = I^2 R$
(B) $P = \frac{V^2}{R}$
(C) $P = VI$
(D) $P = \frac{R}{V^2}$

40. Which of the following statements is true of a secondary cell?

(A) It is formed by connecting two or more primary cells.
(B) It can be recharged by passing a direct current in the opposite direction to which it delivers current.
(C) It can be recharged by passing a direct current in the same direction as it delivers current.
(D) It can be recharged by 'topping up' with dilute sulphuric acid.

Item 41 refers to the diagram below which shows a battery connected across three resistors connected in parallel.



41. It is ALWAYS true that the

(A) potential difference is the same across each resistor
(B) current is the same in each resistor
(C) applied voltage is proportional to the net resistance
(D) voltage across a given resistance is dependent on the value of the resistors

42. Which of the following gives the correct colour code for electrical wiring?

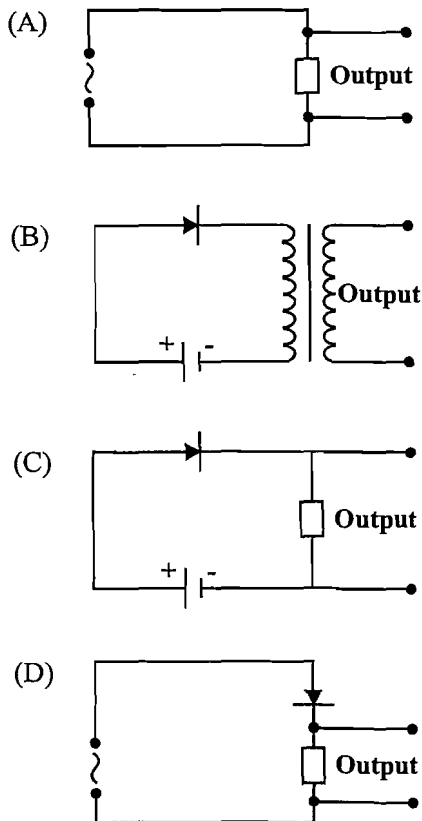
LIVE NEUTRAL EARTH

- (A) Brown Blue Green/
yellow
(B) Red Blue Green
(C) Blue Green/yellow Brown
(D) Green Brown Blue

43. The resistance of an IDEAL ammeter is assumed to be

- (A) zero
(B) 0.1 ohm
(C) 1 to 2 ohm
(D) infinite

44. Which of the following circuits can be used for half-wave rectification?



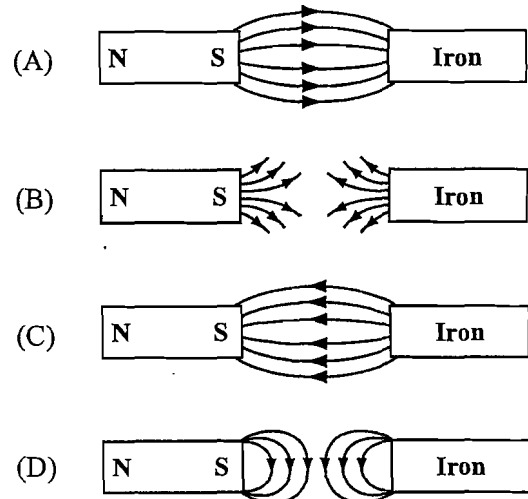
45. A suspended bar magnet, after swinging freely, always settles

- (A) with its poles pointing the same way
(B) with its N pole pointing geographic south
(C) perpendicular to the Earth's magnetic field
(D) opposite to the direction from which it started

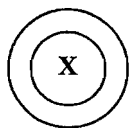
46. Magnetic induction occurs when

- (A) a N pole attracts a S pole
(B) iron nails near a magnet become magnetised
(C) a magnet is suspended and points in the NS direction
(D) an electroscope is charged

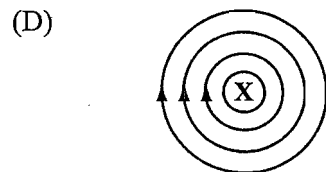
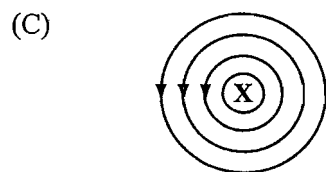
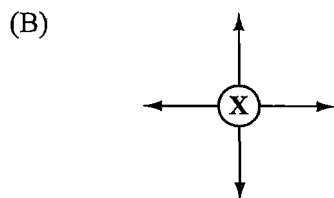
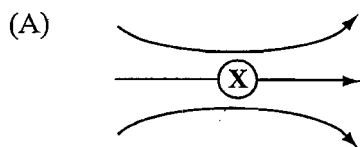
47. Which of the following shows the magnetic field between a bar magnet and a piece of iron?



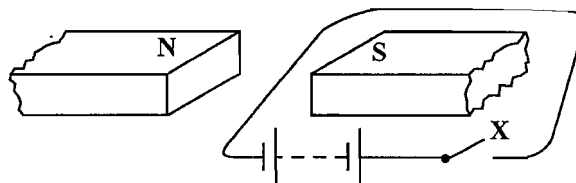
Item 48 refers to the diagram below which represents a straight wire carrying a current into the plane of the paper.



48. Which of the following diagrams BEST represents the magnetic field around the wire?



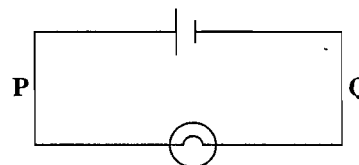
Item 49 refers to the following diagram.



49. The flexible wire shown in the diagram above is loosely supported in the strong magnetic field. When the switch, X, is closed the wire moves

- (A) downwards
- (B) upwards
- (C) towards the north pole
- (D) towards the south pole

Item 50 refers to the following diagram.



50. The current in wire P is 0.9 A. The current in wire Q is

- (A) 0 A
- (B) 0.8 A
- (C) 0.9 A
- (D) 1.0 A

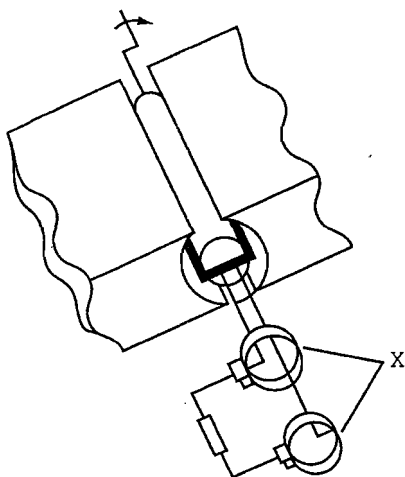
51. A conductor, rotating in a uniform magnetic field, induces maximum instantaneous current when the conductor cuts the magnetic field lines at

- (A) 30°
- (B) 45°
- (C) 90°
- (D) 180°

52. The device for changing the voltage of an alternative supply is called a

(A) transformer
(B) transducer
(C) transistor
(D) transposer

Item 53 refers to the following diagram of a simple a.c. generator.



53. The parts labelled X in the diagram are known as the

(A) coils
(B) armatures
(C) slip rings
(D) commutators

54. In the scattering experiment conducted by Geiger and Marsden, some of the alpha particles were deflected. The explanation for this phenomenon is that

(A) electrons have a small mass
(B) electrons have a small charge
(C) the metal foil was only a few atoms thick
(D) the nuclear charge and mass are concentrated in a small volume

Item 55 refers to the following table.

Nuclei	Atomic Mass	Neutron Number
P	16	8
Q	13	9
R	18	10
S	21	11

55. Which pair of nuclei are isotopes?

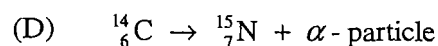
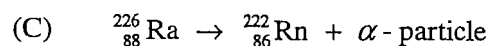
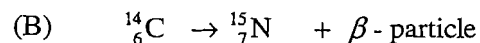
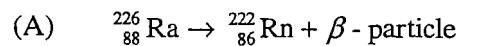
(A) P and Q
(B) Q and R
(C) P and R
(D) Q and S

56. When alpha particles, beta particles and gamma rays from a radioactive source are subjected to a transverse magnetic field, which of them would NOT be deviated from their original path?

I. Alpha
II. Beta
III. Gamma

(A) II only
(B) III only
(C) I and II only
(D) II and III only

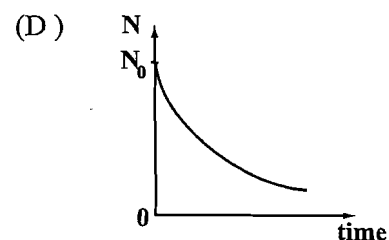
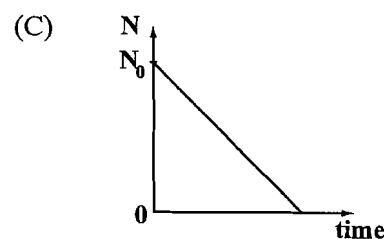
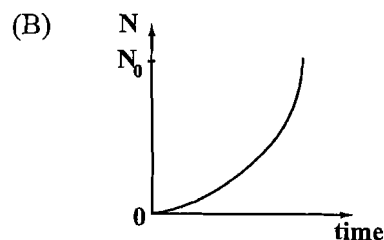
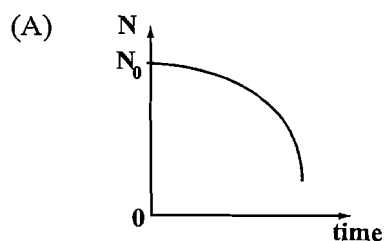
57. Which of the following equations for nuclear reactions is correct?



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58. N_0 radioactive nuclei are present in a sample at time $t = 0$.

Which of the following graphs BEST represents the variation with time of the number, N , of undecayed nuclei present?



59. Which of the following are definitions of the term 'half-life' of radioactive nuclide?

- I. The time taken for the activity of any give sample to fall to half its original value.
- II. The time taken for half the nuclei present in any given sample to decay.
- III. Half the average number of disintegrations per second.

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

60. In the equation $\Delta E = \Delta mc^2$

- (A) c = speed of light, Δm = mass of atom
- (B) c = speed of light, Δm = mass lost
- (C) c = specific heat capacity of substance, Δm = mass lost
- (D) c = specific heat capacity of substance, Δm = mass of substance

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.